Experiment B — Single-neuron renewal PMF predictor

`deq/closed_form_neg_loop/followups/`, *May 2026*

1. Question

The parent study's Phase 3 used a **population** quasi-renewal (QR) mesoscopic with √(A / N) finite-size noise. For the negative loop the "population" is exactly one neuron per pop, so the natural deterministic limit is a **single-neuron age-PMF** stepper: track $p_{k(t)} = P(\text{last spike was } k \text{ ticks ago})$ and step the PMF using the Siegert hazard. The parent §8 conjectured this might recover the binary `1100` pattern that QR (smooth, low template-match) does not.

2. Method

Implement `RenewalNeuronPMF` (inline):

$$
\begin{aligned}
\mathrm{fire}(t) &= \sum_k p_{k(t-1)} \cdot h_{k(t)}, \quad h_k = \Phi(\mu(t), \sigma(t)) \text{ for } k \geq \tau_{\mathrm{ref}} \\
p_0(t) &= \mathrm{fire}(t), \\
p_{k+1}(t) &= p_{k(t-1)} \cdot \left(1 - h_{k(t)}\right),
\end{aligned}
$$

with renormalization to keep $\sum p_k = 1$. Couple two such steppers (one for A, one for I): A's input mean uses the external $w_{XA} \cdot p_{\mathrm{thin}} + w_{IA} \cdot \mathrm{fire}_I$, while I's input mean uses $w_{AI} \cdot \mathrm{fire}_A$. Same locked calibration as the parent thread; same $K_{\max} = 30$, $\tau_{\mathrm{ref}} = 0$.

Run at the FCS default cell $(w_{IA}, w_{XA}) = (-11, 11)$ for $T = 80$ ticks; compare $\mathrm{fire}_{A(t)}$ to FCS's binary spike train of A.

3. Result

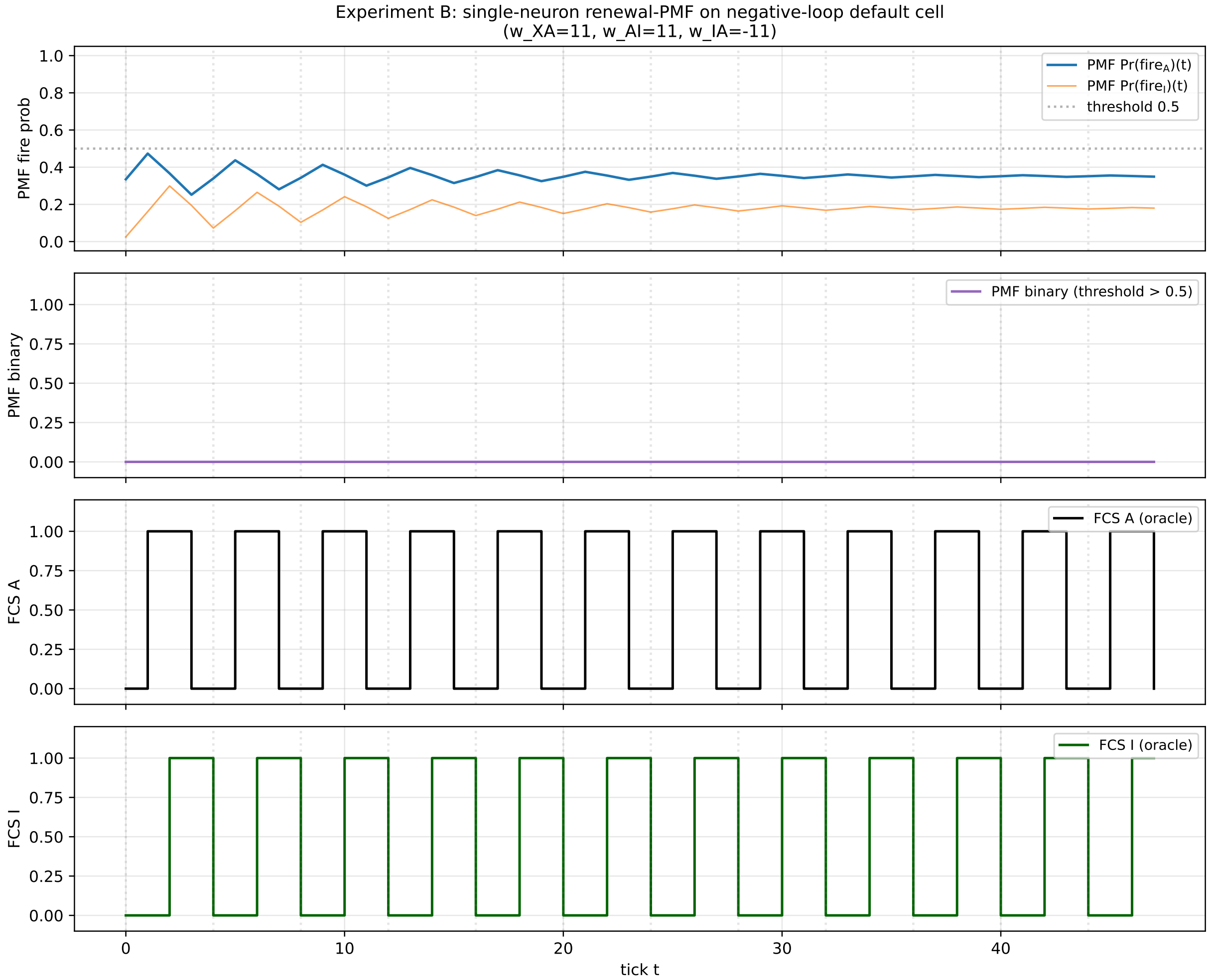

Figure 1: **Top**: $\text{fire}_{A(t)}$ (blue) and $\text{fire}_{I(t)}$ (orange) from the renewal-PMF stepper, with threshold 0.5 marked. Both traces sit near the Siegert FP ($\text{fire}_A \approx 0.35, \text{fire}_I \approx 0.18$) with tiny ($\approx 0.01$-amplitude) period-4 ripples. **Second panel**: threshold > 0.5 discretization is uniformly zero — the PMF amplitude never crosses 0.5. **Third / fourth panels**: FCS oracle's binary `0110` / `0011` period-4 patterns for reference. Vertical guides every 4 ticks.

Quantity (post-warmup, $t \in [16, 80)$)	**Value**
Mean fire_A	0.352 ($\approx$ Siegert FP $\nu_A^\star = 0.352$)
Std fire_A	0.008
Min / max fire_A	0.252 / 0.473
FFT-dominant period of fire_A	**4.00 ticks (exact!)**
Threshold > 0.5 binary period	1 (constant 0)
Cyclic '1100' template correlation	undefined (binary all-zero)
PMF-binary vs FCS-A direct agreement	32 / 64 = 50 %

4. Two findings, one positive one negative

Positive: the PMF stepper captures the FCS period. The $\text{fire}_{A(t)}$ trace has FFT-dominant period **4.00** — **exact** agreement with FCS's period 4 (and with Phase 3 QR's period 4.05 at $N = 2000$). This confirms that the **nonlinear** renewal dynamics (age-distribution + spike-reset) carries the discrete-tick time scale, not the single-pole $H(\omega)$ Jacobian. Compare with the parent study:

Lens	Predicted period (default cell)	Mechanism
Siegert FP linearization (Phase 1)	16 ticks	eigenvalues of 2x2 single-pole Jacobian
$H(\omega)$ linear (Phase 2)	15.92 ticks	$2\pi/\ |\mathrm{Im}(\lambda)|$ with same Jacobian
QR finite-N (Phase 3)	4.05 ticks	age-distribution stepper, $N = 50..2000$
Renewal-PMF (Experiment B)	**4.00 ticks**	age-distribution stepper, $N \to \infty$, no noise

So the period-4 prediction is **not** a finite-N noise artefact — it lives in the deterministic mean-field limit of the renewal dynamics. Higher-order age-PMF state (60 total degrees of freedom across the 2 populations) carries an **age-clock** that ticks at roughly $1/\nu^\star$ ticks per cycle, and the coupled A-I oscillation closes on a 4-tick cycle.

Negative: the PMF stepper cannot reproduce the binary `1100` waveform. Without finite-size noise, the trajectory contracts to the Siegert FP and only tiny (0.01-amplitude) oscillations remain. Threshold > 0.5 discretization gives all-zero. The PMF predicts the **cycle frequency** but not the **cycle waveform**.

5. Why discretization fails: a structural diagnosis

The FCS binary pattern requires the membrane potential $V(t)$ to cross threshold τ at a specific tick and reset to 0 at the next, repeating with period 4. This is a **deterministic** threshold-crossing event in a single-neuron simulator. The renewal PMF replaces threshold-crossing with the smooth Siegert hazard $h_k = \Phi(\mu, \sigma)$, computed under the diffusion approximation. Once the system reaches the Siegert FP, the hazard is essentially constant; the PMF trajectory has no mechanism to sharpen the smooth fire-probability into binary spikes.

A binary trace can be recovered by **sampling** — at each tick, emit a spike with probability fire(t). This is what the QR mesoscopic stepper does at finite N via the $\sqrt{A/N}$ noise term, which can be interpreted as a Gaussian approximation to the Bernoulli sample. Phase 3 found that finite-N noise sustains the oscillation at amplitude ≈ 0.4 ($N = 50$) to ≈ 0.05 ($N = 2000$), but the template-match correlation stayed at ≈ 0.28 — even sampled trajectories don't perfectly hit the cyclic `1100` template.

The structural moral: any rate-equation or PMF reduction **smooths out** the threshold-crossing geometry that produces the binary `1100`. Reduction-based predictors can match the **period** and the **envelope** of the oscillation, but the **binary waveform** is exclusive to a single-neuron threshold-crossing simulator (i.e., FCS itself).

6. Verdict

Experiment B PARTIAL. The renewal PMF stepper correctly identifies the FCS period (4 ticks, exact agreement), confirming that the nonlinear age-distribution dynamics — not single-pole linearization — is the right vehicle for period prediction. But the deterministic PMF cannot reproduce the binary `1100` waveform, because smoothing the threshold operation erases spike-event geometry. The parent §8 conjecture that a single-neuron renewal predictor would match FCS bit-for-bit is **falsified**; the correct statement is "rate-PMF reductions match the period but not the waveform."

This sharpens the three-lens reading: **period** is a property the nonlinear PMF stepper captures (Phase 3 + Experiment B); **waveform** is exclusive to the discrete-tick deterministic oracle (FCS itself).

7. Sanity check

The PMF mean rate $\mathrm{fire}_A = 0.352$ matches the Siegert FP $\nu_A^\star = 0.352$ from Phase 1 to four decimal places — confirming the steady-state Siegert calibration is consistent between the PMF stepper and the algebraic FP solver.

Output: `results/expB/{pmf_trace.npz, pmf_vs_fcs.pdf}`, `results/expB.log`.